

Top layer: 56 named wrappers

`osc_prob_{2nu, 3nu, 4nu, 5nu}_...`

Standard

`...vacuum`
`...matter_constant_density`
`...matter_exp_density`
`...earth`
`...sun`

Non-standard interactions

`...matter_nsi_constant_density`
`...matter_nsi_exp_density`
`...earth_nsi`
`...sun_nsi`

Lorentz-invariance violation

`...vacuum_liv`
`...matter_liv_constant_density`
`...matter_liv_exp_density`
`...earth_liv`
`...sun_liv`

Parameter
dictionary

Second layer: scenarios

`osc_prob_vacuum`
`osc_prob_matter_std_potential`
`osc_prob_matter_nsi`
`osc_prob_liv`

H, if
no engine
applies

Third layer: scan

`osc_prob_energy_baseline`
Engine 6 if it applies;
else one call per point,
with warm starts.

One point
at a time

Base layer

`osc_prob`
Engine 7, the general
Magnus ladder: reached
when no other applies.

Tried first

Engines 1–5, tried in order

- 1 Averaged probability
- 2 Adiabatic + Magnus patches
- 3 Interaction picture
- 4 Constant Hamiltonian
- 5 Energy-batched scan

Tried first

Engine 6

Cumulative scan
over baselines